DEMONSTRATING ADS-B AND CPDLC ATTACKS WITH SOFTWARE-DEFINED RADIO

Sofie Eskilsson, Hanna Gustafsson, Suleman Khan, Andrei Gurtov
Department of Computer and Information Science, Linköping University, Sweden

Abstract

Several studies have shown insufficient security in air traffic communication. Controller-Pilot Datalink Communications (CPDLC) is used to communicate in text over the VHF data link, and Automatic Dependent Surveillance – Broadcast (ADS-B) determines the position of an aircraft. The vulnerability of air data communication was confirmed by successful experiments using Software-Defined Radio, where both CPDLC and ADS-B messages were transmitted in a safe environment. Neither ADS-B messages nor CPDLC messages are encrypted during transmission. The encoding of FANS-1/A messages was demonstrated, and the experiments showed that it is possible to send such messages with relatively inexpensive technology.

I. Introduction

Air Traffic Control (ATC) is the backbone of the world's air transport, and as traffic load continues to grow dramatically, ATC has to handle more and more aircraft. A new technique has become necessary to unburden the voice communication. As a standard method of communicating between ATC and pilot, the voice radio, Very High Frequency (VHF), is used for short-distance communication. As a solution to the high demand in air traffic communication, data link-based communication was introduced. By increasing the sufficient capacity of the communication channel, the technology can be used more efficiently.

This technique, called Controller Pilot Data Link Communications (CPDLC), complements VHF radio communication, and take care of the non-critical communication. Since the introduction, it has reduced accidents and increased communication effectiveness.

To determine an aircraft's position, ground-based radar was previously used. Due to the extraordinary growth rate of the aircraft traffic volume [1], the limits of the current system were approached, and the implementation of a new one was crucial. The technology developed, Automatic Dependent Surveillance-Broadcast (ADS-B), utilizes the Global Positioning System (GPS) and the Global Navigation Satellite System (GNSS) to determine the exact coordinates of the aircraft. The coordinates are then automatically transmitted via air-to-ground and air-to-air data communication links at a fixed rate, depending on the aircraft's current phase.

Like any new technological advancement, security may not always be at the forefront of the system's development, something we will take a closer look at. In this paper, we will demonstrate how to simulate ADS-B and CPDLC messages to investigate the vulnerability in aircraft communication. We will utilize HackRF One to transmit CPDLC and ADS-B messages at insignificant power in a closed environment for testing. An RTL-SDR dongle with an antenna will be used to capture the messages.

The sections below will be structured as follows. A brief overview of related work, followed by concepts and theories concerning ADS-B, CPDLC and a description of the equipment. The remaining sections describe the simulation of ADS-B and CPDLC messages and how they were encoded, transmitted, received, and decoded.

II. Related Work

Several studies discuss the uncertainty in aviation communication. In [2], Gurtov et al. highlight the lack of security regarding CPDLC. In [3], McCallie et al. analyzed the security vulnerability associated with ADS-B and executed a security analysis. Sestorp and Lehto [4] show that it is possible to capture and decode CPDLC messages. All these papers discuss the possibility to possess cheap radio equipment, such as

978-1-7281-7270-5/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

SDR, to access and monitor data link communications and decode these.

III. Background

ATC protocols enable communication between controllers and pilots. The primary purpose of an ATC is to avoid all kinds of aircraft collisions, organize the flow of air traffic, and provide support for pilots [2]. ATC is a type of Aircraft Communications Addressing and Reporting System (ACARS) which is a digital data link system for the transmission of messages between aircraft and ground stations. Voice communication is the primary means of communication between ATC and an aircraft. It is used to transmit all ATC instructions to the aircraft, which are acknowledged by the pilot, as well as pilots' reports and requests to ATC. Flight information services, weather reports, and airport information broadcasts can also be provided by voice communication. It is further used for operational communication between the airline operator and the aircraft, as far as the aircraft is in the range of the operator's transmitter. Voice communication is conducted by analog radio on VHF [5].

The Automatic Dependent Surveillance-Broadcast (ADS-B) technology is the mainly used in today's aircraft for communication between aircraft and the controllers working in ATC. The aircraft determines its position via satellite navigation and periodically broadcasts it, enabling it to be tracked. This system is used to improve many factors for both pilots and air traffic controllers, including collision avoidance and situational awareness. While ADS-B improves communication, it also exhibits weaknesses, mainly that ADS-B is not encrypted and that any person with the correct gear can both read and send ADS-B data without any difficulty, which calls for improvement [3].

Controller-Pilot Data Link Communications (CPDLC) is a message-based communication between ATC and aircraft. ATC can use CPDLC via a terminal to send clearances or requests. The aircraft can choose between several predefined phrases or free texts to send to the ATC. This technique has several advantages over voice radio, such as the reduced risk of acoustic misunderstandings. It is also more comfortable and more efficient to transmit and receive longer messages. The voice radio is dependent on the controller handling many operations simultaneously, while that is not a problem using CPDLC. Some of the busiest airports, and all the European, already employ CPDLC for automated clearance delivery and start-up approval [6]. The system includes airspaces not covered by VHF and ensures availability, even in oceanic airspace, and enables the possibility of connecting through a satellite communications network.

CPDLC is currently available in two formats: Future Air Navigation System (FANS-1/A) and Aeronautical Telecommunications Network - Baseline 1 (ATN-B1). FANS-1/A is, despite its name, the old technology and used widely across the world. It is the only CPDLC format that can be carried over any aviation radio link, capable of carrying ACARS messages seen in Figure 4. ATN-B1 is the new format that is now in operation only in Europe. This format is only supported on the VDL2 radio link [7].

IV. Experimental Testbed

This section presents the equipment that was used in the experiments and an associated validation. The remainder of this section describes a general approach to encode, transmit, receive and decode ADS-B and FANS-1/A messages.

A. Setup and Validation of Equipment

The experimental setup, seen in Figure 1, includes:

- Computers
- HackRF One
- RTL-SDR dongle (R820T2 RTL2832U)
- Antenna 500
- Antenna (VHF)



Figure 1. Picture of the Hardware

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

HackRF One is hardware for Software-Defined Radio (SDR). It is a technology capable of transmitting and receiving radio signals, where HackRF One was used to transmitting signals in the experiments. Unlike traditional radio communication, where the components such as modulators and tuners, are implemented in analog hardware, components in SDR were implemented in software. With SDR, the software makes it possible to create programmed solutions [8]. HackRF One connects to the computer via USB and allows signals from the frequency between 1 MHz to 6 GHz. It provides a software-controlled antenna port power of 50 mA at 3.3 V and an output power of 16.5 mW [9].

Realtek-Software Defined Radio (RTL-SDR) is a USB dongle used for receiving live radio signals. The RTL-SDR v.3 is based on the RTL2832U chipset with an R820T2 tuner. It can receive frequencies between 500 kHz – 1766 MHz with bandwidth up to 2.4 MHz [10].

In ADS-B technology, two different types of equipment are distinguished; *ADS-B Out* and *ADS-B In*. The Out component takes information from the GPS and other equipment and broadcasts the information to ground stations or other aircraft in proximity. When an ADS-B signal has been received, any receiver in the transmission range with ADS-B In equipment gets access to the flight number, altitude, velocity, latitude, and longitude [11]. In the experimental setup, RTL-SDR dongle corresponds to the ADS-B In equipment.

By setting up the RTL-SDR dongle with the antenna, it was verified that ADS-B messages from several aircraft passing by could be captured. Thus, the decoding software could be validated by comparing the captured ADS-B messages with the ADS-B data from the online flight tracking service Flightradar24 [1]. Figure 3 displays the interface of the renowned website Flightradar24 that was used to scout suitable aircraft in real-time, here showing flight *FIN7WG* heading towards London, accompanied by ADS-B data seen bottom left in the figure. As can be seen in Figure 3, data from Flightradar24 match the data derived with the decoder, ADSB-out. The first field *Hex* is the airport code (ICAO-code), *Track*

[1]https://www.flightradar24.com/

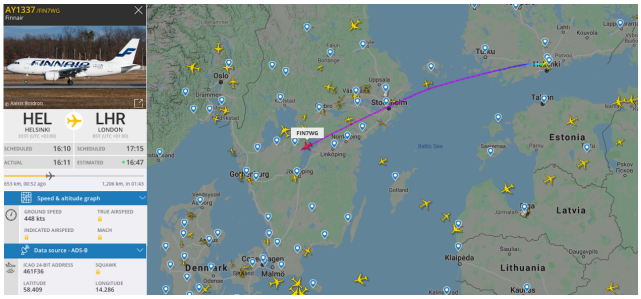


Figure 2. Graphical View of Flight "FIN7WG" in Flightradar24

Hex	Flight	Altitude	Speed	Lat	Lon	Track	Messages	Seen
461f36	FIN7WG	37950	448	58.408	14.281	243	517	0 sec

Figure 3. Decoded ADS-B Message in the Terminal from Real-time Aircraft

is the angle of the flight, *Messages* is the number of messages received and finally *Seen* is the time at which the last packet was received.

B. Simulation Process of ADS-B

ADS-B messages use a transmission frequency of 1090 MHz, a surveillance transmission mode called Mode S. Mode S transponders will integrate a message called Extended Squitter (ES). Each 1090ES-message has a total duration of 120 µs, consists of 112 bits, and is sent at least once a second. Each message is divided into five fields: Downlink format, Capability, Aircraft Address, ADS-B Data, Parity Check seen in Table I [12].

Table I. F.ormat of a 1090 Extended Squitter Message

Segment	Downlink Format	Capability	Aircraft Address	ADS-B Data	Parity Check
Bits	5	3	24	56	24

The first field *Downlink Format* can be in different formats depending on what type of message it is. For example, a squitter message uses format 17. The second field *Capability* shows the communication capability of the Mode S transponder, and the *Aircraft Address* identifies the sender with a unique 24 bits address. *ADS-B Data* consists of further segments shown in Table II and in *Parity check* any possible transmission error will appear [3].

Before transmitting ADS-B data in Table II, the message needs to be encoded to simulate a real

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

Table II. Example of ADS-B Data of Airborne Position

ADS-B Data								
Segment	TC	SS	SAF	ALT	T	F	LAT-CPR	LON-CPR
Bits	5	2	1	12	1	1	17	17

ADS-B message. The encoded message can easily be transmitted using HackRF One and received with the RTL-SDR dongle. The last step, to read the received message, requires a decoding process.

C. Simulation Process of FANS-1/A

To create a FANS-1/A message, the format must be specified and structured in the Abstract Syntax Notation One (ASN.1). The syntax is described in [13], which defines the structure of the message. A FANS-1/A shall use ISO/IEC 8825-2:1996 Packed Encoding Rules (PER) - Basic Unaligned [14]. To transmit the PER encoded FANS-1/A message as an ACARS message, it must be encapsulated into an ARINC-622 Air Traffic Service (ATS) message.

<ACARS address> <IMI> <Aircraft Registration Number> <Byte String> <CRC>

Figure 4. ARINC-622 ATS Message Structure

The ACARS address is a specific 7-character address based on the 4-character airport code (ICAO-code), unique for each aircraft. The Imbedded Message Identifier (IMI) shows what kind of connection that is established [14], [15]. The following seven characters are the aircraft registration number; the first one, two or three characters, refer to the origin nation, and the subsequent characters are a unique combination within the nation [7]. The next part of the message is the PER-encoded FANS-1/A message as a byte string.

To generate the byte string, ASN.1 Studio [2] was used. ASN.1 Studio is an integrated development environment (IDE) used to create ASN.1 values graphically. The environment generates code from the ASN.1 specification and includes various protocol data units (PDU) such as *ATCdownlinkmessage*, *ATCdownlinkmsgelementid*, and *ATCmessageheader*. The *ATCdownlinkmessage* is a sequence of *ATCmessageheader* and *ATCdownlinkmsgelementid*. Up-link message refers to communication going from ground to air, and a down-link message refers to communication going from air to ground. Every down-link and up-link messages contain an element ID, which represents a unique type of message; examples of down-link messages can be seen in Table III.

Table III. Examples of Down-link Messages

ID	Message element	Resp
0	WILCO	N
1	UNABLE	N
2	STANDBY	N
3	ROGER	N
4	AFFIRM	N
5	NEGATIVE	N
9	REQUEST CLIMB TO [altitude]	Y
29	CLIMBING TO [altitude]	N
56	MAYDAY MAYDAY MAYDAY	N
67	Free Text	N
68	Free Text	Y

The last four characters in Figure 4 are hex digits calculated by a Cyclic Redundancy Check (CRC), used to detect errors in data storage or transmission. Blocks of data entering these systems get a short check value attached, based on the remainder of a polynomial division of their contents. This simple detection mechanism works if an odd number of bits in a byte changes, but an even number of false bits in one byte will not be detected by the parity check. The CRC in ARINC - 622 ATS message is called a 16-bit CRC with polynomial coefficients of $x^{16}+x^{12}+x^5+1$ [16]. The CRC was calculated by using the IMI, aircraft registration number, and the PER-encoded byte string. Parts of an open-source project *Libacars* [3] was also used. The method used to generate a correct CRC can be seen in Figue 5.

HackRF One, with an antenna, was utilized to transmit FANS-1/A messages. Once the message had been modified into the correct format, GNU Radio was used to generate a signal to enable the message to be sent through HackRF One. A text file consisting of the encoded FANS-1/A message was added into the software, where the component *GMSK Mod* transform the byte stream input to a complex modulated signal. The output was a complex signal sent with HackRF One.

The antenna connected to the RTL-SDR dongle received the signal. GNU Radio managed to transform

[2]http://www.oss.com/asn1/products/asn1-products.html

[3]https://github.com/szpajder/libacars

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

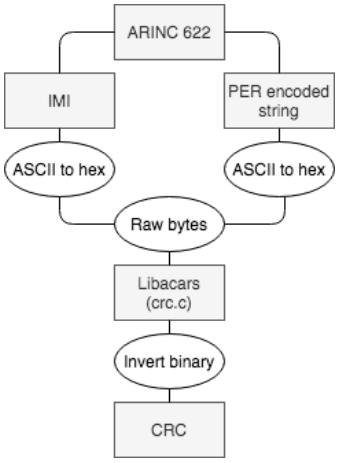


Figure 5. Schematic of CRC Creation

a complex signal and generating an output of a bitstream with the component *GMSK Demod*. The received message was saved in a file.

The software *Libacars* made it possible to decode the FANS-1/A message. The decoding involved verifying the CRC, decoding the PER-encoded string, and finally pass it into the correct format using ASN.1. The FANS-1/A message can now be read by the receiver and have the same format and information as the message sent. The entire process step by step is shown in Figure 6.

V. Performance Evaluation

This section presents the execution, step by step, when transmitting and receiving a selected ADS-B and FANS-1/A message, by using the simulation processes in IV.B, IV.C.

A. ADS-B

To create and encode a message called test_sample, an open-source project was utilized called *ADSB-Out*[4]. The desired signal was created by running the script ADS-B_Encoder.py where NameOfFile was added to make it possible to save several signals with different values. The script was executed with the arguments: *ICAO, Latitude, Longitude, Altitude, NameOfFile.*

The example signal named *test_sample* was created including following information; ICAO *abc123*,

[4]https://github.com/lyusupov/ADSB-Out

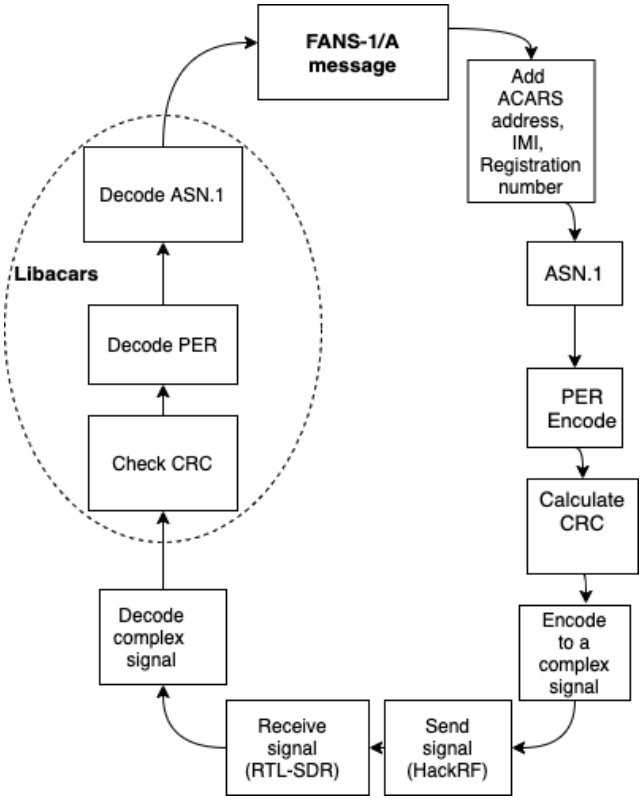


Figure 6. Implementation of FANS-1/A

latitude *15.575702*, longitude *58.397548* and altitude *975.0*. The created file *test_sample.iqs* was further aligned to 256K buffer size. A execution of ADSB_Encoder.py looks as follows:

```
$ cd Documents/ADSB-out
$ python ADSB_Encoder.py 0xabc123
 ↪  58.397548 15.575702 975.0
 ↪  test_sample
$ dd if=test_sample.iq8s
 ↪  of=test_sample_256K.iq8s bs=4k
 ↪  seek=63
```

The created signal was encoded with *ADSB-Out* and transmitted recursive with HackRF One over the licence free frequency 868 MHz using sample frequency 2MHz.

```
$ cd Documents/ADSB-out
$ hackrf_transfer -t
 ↪  test_sample_256K.iq8s -f
 ↪  868000000 -s 2000000 -x 47 -R
```


Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

RTL-SDR dongle received the signal, and to decode the example signal; an open-source project was used called *dump1090* [5]. *Dump1090* is an ADS-B decoding program that is compatible with the RTL-SDR dongle.

To receive and decode the example signal, *test_sample*, the frequency to capture was set to 868 MHz since it was the chosen sending frequency. The decoded message can be seen in Figure 7.

```
$ cd Documents/ADSB-out
$ sudo ./dump1090/dump1090 --net
  ↪  --freq 868000000
```

```
Hex    Flight   Altitude  Speed  Lat       Lon       Track  Messages Seen    .
--------------------------------------------------------------------------------
abc123          975       0      58.398    15.576    0      13214     0 sec
```

Figure 7. Received and Decoded ADS-B Message

B. FANS-1/A

In the following section, all steps required to encode, transmit, receive, and decode a staged example FANS-1/A message, carrying the free text message *TEST TEST TEST* are presented.

1) Message Creation

Recall the message structure from Figure 4; the ACARS address was set to AKLCDYA. In this case, the ATS was situated in Auckland. The IMI was set to AT1 to declare a CPDLC message, in an already established contact. The seven characters long registration number of the aircraft was set to .9M-MTB. The following string carried the message element of the signal, and the last four digits were the CRC calculated from the test message.

In ASN.1 Studio the free text message *TEST TEST TEST* corresponds to ID 67 in the *ATCDownlinkMessage* format, as seen in Table III. The program thus returned the string *002186D48B4EA20A9169D441522D3A80*, see Figure 8, which is the *TEST TEST TEST* message formatted as an ATC Downlink Message and encoded with Unaligned Packed Encoding Rules (PER).

The last thing was to calculate a CRC to have a complete message. The first steps of generating the CRC can be seen in the transformations in Table IV.

[5]https://github.com/antirez/dump1090

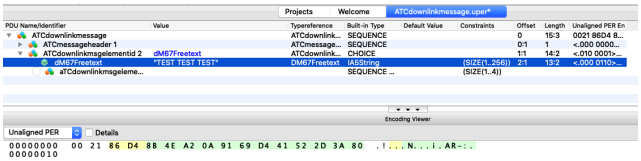


Figure 8. The Software ASN.1 Studio

Table IV. Calculating CRC

	Text	Hex	Raw bytes
IMI	AT1.9M-MTB	4154312E39 4D2D4D5442	0x41, 0x54, 0x31, 0x2E, 0x39, 0x4D, 0x2D, 0x4D, 0x54, 0x42
PER String	TEST TEST TEST	002186D48B4 EA20A9169D4 41522D3A80	0x00, 0x21, 0x86, 0xD4, 0x8B, 0x4E, 0xA2, 0x0A, 0x91, 0x69, 0xD4, 0x41, 0x52, 0x2D, 0x3A, 0x80

Merging the raw bytes of the IMI and the PER-encoded string, run it in the modified version of *Libacars*, returned the string 320D. Inverting the binary string of 320D yield CDF2. The complete test message was now ready to be transmitted in the following format:

AKLCDYA. | AT1 | .9M-MTB | 002186D48B4EA20A9169D441522D3A80 | CDF2

Figure 9. Complete FANS-1/A Message

2) Message Transmission

Using HackRF One, the message was transmitted as a complex signal. To transform the signal, the instrument GNU Radio used different modules to convert the signal. A file containing the correct format of the message was imported to GNU Radio, see Figure 10. The signal then went into the Packet encoder where it got wrapped into a packet with a header, access code, and preamble. The signal then passed through a block of Gaussian Minimum Shift Key (GMSK) modulation. In the last operation, the complex signal passed to the Osmocom Sink, which sends with HackRF One.

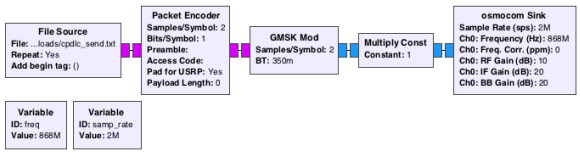


Figure 10. Transmitting a FANS-1/A Message with GNU Radio

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

*3) **Message Decoding***

The signal was successfully received by the RTL-SDR dongle within a distance of approximately one meter and brought into GNU Radio once again. Figure 11 shows how the RTL-SDR source block picks up the signal and is specifically tailored for the use of the RTL-SDR-dongle.

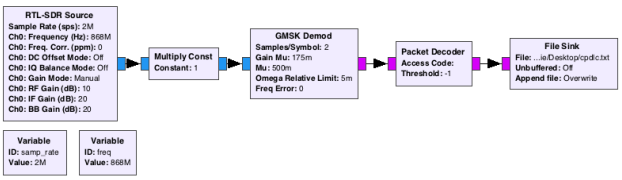


Figure 11. Receiving a FANS-1/A Message with GNU Radio

The message is now in the phase of decoding, to see if the message element carries the message *TEST TEST TEST*. The first row expresses that this is a down-link message (d) with the label AA. The label is used to identify the application carried in the message text.

```
$ decode_acars_apps d AA
 ↪  /AKLCDYA.AT1.9M-MTB002186D48B4E
 ↪  A20A9169D441522D3A80CDF2

/AKLCDYA.AT1.9M-MTB002186D48B4EA20
A9169D441522D3A80CDF2
FANS-1/A CPDLC Message:
 CPDLC Downlink Message:
  Header:
   Msg ID: 0
  Message data:
   [freetext]
    TEST TEST TEST
```

This is the end operation and the decoded message returned *TEST TEST TEST*.

VI. Result and Analysis

This section is divided into the following subheadings; ADS-B and FANS-1/A. In each subheading, the result and the analysis will be discussed.

A. ADS-B

The implementation to send and receive an ADS-B signal was relatively painless. An encoder and decoder were found quickly, and because of the easy way to monitor a real aircraft, it was easy to verify that the decoder worked as intended. The thorough documentation in the subject helped during the execution of the experiments. Due to the simple implementation, together with the systematic documentation, this increases the risk of a possible attack. The relatively inexpensive equipment can also result in an increasing number of people being able to carry out an attack, and the acquisition of more devices can lead to a large-scale attack. The fact that ADS-B does not require a logon is an additional security flaw.

When the setup initially had been verified, a maximum range for sending the signal was tested. The range test was executed in an open field without any distraction to disturb the signal. The measured distance for the signal is approximately 50 meters, which means that an attacker can easily inject messages into the ATC system or nearby aircraft. With an amplifier, an attack could be executed from even further distance. The staged aircraft in the experiments was standing still with no speed or angle. To make the attack more realistic, for future work, a script can, with ease, be made that stage a moving aircraft.

Delimitation in the experiments was the frequency. Due to regulations, a license is required to transmit over the frequency of 1090 MHz to avoid interference for air traffic. The frequency used for the experiments was chosen to 868 MHz after some consideration. The choice was between the frequency 433 MHz and 868 MHz, both reserved for Short Range Devices, and are license-free all over EU. 868 MHz is characterized by the power efficiency and ranges from a couple of meters, up to several kilometers. The 868 MHz band is more or less interference-free, with great damping in contrast to 433 MHz, which is more cluttered.

B. FANS-1/A

During the research, a problem was experienced to find a good way to encode CPDLC messages. There was software to decode a FANS-1/A message, but none to encode. A reason for this could be that the hardware to encode and transmit is way more expensive than the hardware to decode and receive a message. A consequence of this could be that less software for encoding has been developed. This can

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

contribute to making it harder to perform an attack since no complete program for encoding exists.

As mentioned in section V.B-3, the signal was received at about one-meter distance from the sender. The signal was not successfully received in every attempt because the connection was quite unstable. The hardware's, HackRF One and RTL-SDR-dongle, are capable to transmit and receive on a distance of at least 50 meters. For future work, it is possible to modify and optimize the signal sent with GNU Radio. If a more stable signal were created, it could have been transmitted at a greater distance. In the test executed, it is shown that HackRF One can be utilized to send a file containing a FANS-1/A message. For further work, a script can be created to avoid sending the message over a file.

The section V.B-3 confirmed that the encoding had been done correctly. The desired message, in this case, *TEST TEST TEST*, could be encoded, using the method developed, and then decoded and returned to the recipient. In the test, pre-defined, and free text messages can be sent in a file. But if a real aircraft receives any CPDLC messages without being connected, the aircraft will ignore them. This means that a connection needs to be established, to simulate a real attack. Since CPDLC accepts all requests as long as they contain the correct information, it means that it does not question the sender and an attack is harder to discover.

As mentioned in section I, the focus of this thesis has been to send a message of the format FANS-1/A. The reason why FANS-1/A was chosen over ATNB1 was that we found more documentation regarding the encoding of FANS-1/A. Similar to FANS-1/A, ATN-B1 messages are also PER-encoded ASN.1 data structures, and the method developed could have been partially used even for the ATN-B1 message. However, the abstract syntax of ATN-B1 is slightly different than the syntax of FANS-1/A.

VII. Conclusions and Future Work

By successfully transmitting and capturing ADS-B and CPDLC messages in a safe environment, the experiments indicate that it is possible to inject such messages in practice. The data transmitted and captured contains elements with sensitive information that could be utilized to carry out attacks with severe consequences to air traffic. The vulnerability of air data communication was confirmed by successful experiments using Software Defined Radio where both CPDLC and ADS-B messages were transmitted. Neither ADS-B messages nor CPDLC messages are encrypted when transmission of messages in the airspace takes place. However, the messages are sent encoded and a message decoding is thus required on the receiver side to read the message in plain text. The surveys have shown the existence of security flaws due to a relatively easy method to encode CPDLC and ADS-B messages. Encoding for FANS-1/A messages was implemented, and the experiments show that it is possible to send such messages with relatively inexpensive technology. The experiments also show the delimitation in range for sending messages with inexpensive SDR and the distance for CPDLC was considerably more limited than for ADS-B.

In the future, the L-Band Digital Avionics and Communication System (LDACS) can address the security issues. LDACS has security features in place such as broadcast control channel protection, trust, entity authentication, key negotiation, key derivation, confidentiality, integrity and authenticity protection of messages in transit. To improve the security of the broadcast control channel protection hybrid model using LDACS, Timed-Efficient Stream Loss-tolerant Authentication (TESLA) protocol [17] and digital signatures should be included in the control message of ground station (GS) broadcast. For trust purposes, an existing Public Key Infrastructure (PKI) [18] is used in LDACS. PKI is successfully used in AeroMACS and it is deployed in China. For entity authentication and key negotiation, station to station (STS) protocol [19] is recommended. RMAC-based Extract-and-Expand Key Derivation Function (HKDF) [20] is used to generate arbitrary-length session keys with high entropy for key derivation. To achieve confidentiality, integrity, availability and authenticity AES-256 in Galois Counter Mode (GCM) algorithm is recommended in LDACS for implementing.

References

[1] Statista, Growth in Global Air Traffic, Available at https://www.statista.com/statistics/193533/growth-of-global-air-traffic-passenger-demand/ (17.05.2020).

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.

[2] A. Gurtov, T. Polishchuk, and M. Wernberg, "Controller–Pilot Data Link Communication Security," Sensors, vol. 18, no. 5, p. 1636, 2018.

[3] D. McCallie, J. Butts, and R. Mills, "Security Analysis of the ADS-B Implementation in the Next Generation Air Transportation System," International Journal of Critical Infrastructure Protection, vol. 4, no. 2, pp. 78–87, 2011.

[4] I. Sestorp and A. Lehto, "CPDLC in Practice: a Dissection of the Controller Pilot Data Link Communication Security," Linköping University, Aug. 2019.

[5] ICAO, International Standards and Recommended Practices and Procedures for Air Navigation Services, Annex 10 to the Convention on International Civil Aviation, 6th ed., International Civil Aviation Organization, Oct. 2001.

[6] M. Strohmeier, "Security in Next Generation Air Traffic Communication Networks," PhD thesis, Kellogg College, University of Oxford, Jul. 2016.

[7] ICAO, Global Operational Data Link Document (Gold) Manual, 1st ed., International Civil Aviation Organization, Montréal, Quebec, Canada, 2016.

[8] M. Sruthi, M Abirami, A. Manikkoth, R Gandhiraj, and K. Soman, "Low Cost Digital Transceiver Design for Software Defined Radio Using rtl-sdr," in 2013 International Mutli-Conference on Automation, Computing, Communication, Control and Compressed Sensing (iMac4s), IEEE, 2013, pp. 852–855.

[9] G. S. Gadgets, Hackrf one, Available at https://greatscottgadgets.com/hackrf/one/ (2020/03/26).

[10] RTL-SDR.com, Rtl-sdr blog v3 datasheet, Jun. 2017.

[11] M. Arblaster, Air Traffic Management: Economics, Regulation and Governance. Elsevier, 2018.

[12] J. Sun, "The 1090mhz riddle," An Open-Access Book About Decoding Mode-S and ADS-B Data, 2015.

[13] S. Eskilsson and H. Gustafsson, "Simulating ADS-B and Cpdlc Messages with sdr," Linköping University, Jun. 2020.

[14] D. Allen and et al., "Interoperability Requirements for Ats Applications Using Arinc 622 Data Communication Document," RTCA SC-189/EUROCAE WG-53 Position Paper Safety and Interoperability for Air Traffic Services Communication, Oct. 1998.

[15] ICAO, Asia/Pacific Regional Interface Control Document (ICD) for ATS Interfacility Data Communications (AIDC), 3rd ed., ICAO Asia/Pacific Regional Office, Bangkok, Sep. 2007.

[16] T. V. Ramabadran and S. S. Gaitonde, "A Tutorial on CRC Computations," IEEE Micro, vol. 8, no. 4, pp. 62–75, 1988.

[17] A. Perrig and J. Tygar, "Tesla Broadcast Authentication," in Secure Broadcast Communication, Springer, 2003, pp. 29–53.

[18] B. Crowe, "Proposed Aeromacs PKI Specifcation Is a Model for Global and National Aeronautical Pki Deployments," in 2016 Integrated Communications Navigation and Surveillance (ICNS), IEEE, 2016, pp. 1–19.

[19] S. Blake-Wilson and A. Menezes, "Unknown Key-Share Attacks on the Station-to-Station (STS) Protocol," in International Workshop on Public Key Cryptography, Springer, 1999, pp. 154–170.

[20] H. Krawczyk and P. Eronen, "Hmac-Based Extract-and-Expand Key Derivation Function (HKDF)," RFC 5869, May, Tech. Rep., 2010.

Acknowledgement

This work was supported by Trafikverket and Luftfartsverket under Automation Program II.

Authorized licensed use limited to: Lovely Professional University - Phagwara. Downloaded on October 23,2026 at 20:00:52 UTC from IEEE Xplore. Restrictions apply.